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An interactive teaching evaluation system for preschool education in universities based on machine learning algorithm

Deming Li^{*}

School of Education, Jilin International Studies University, Changchun, 130117, Jilin, China

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ABSTRACT

Interactive teaching is very popular in the field of education, especially in college preschool education teaching. In this study, the experimental performance of the evaluation system based on scientific emotion construction is analyzed by the machine learning algorithm. The experimental results show that the classification model realized by the machine learning algorithm is feasible and effective for the construction of an interactive design teaching evaluation system of preschool education in colleges and universities. The main research content of this experiment is for the teaching evaluation text, this paper intends to use NB, KNN, LR, and SVM algorithms as the meta-learning algorithm in the framework, the classification effect of the meta-classifier is ranked as SVM > LR > KNN > NB. From the above data comparison, the SVM meta-classifier in the algorithm in this paper has achieved the best classification effect. The SVM meta-classifier based on the algorithm in this paper performs well, and the classification accuracy rate reaches 92.1%. This shows that this model is very suitable for studying the. And the classification accuracy rate reaches 92.1%. This shows that this model is very suitable for studying the construction of interactive teaching evaluation systems under emotional education.

1. Introduction

Since now the teaching center has shifted the subject from teachers to students, and gradually, teachers and students also have the concept of equal treatment, the idea of interactive design teaching is proposed, which is essentially a scaffolding teaching method. Interactive design teaching is essentially a teaching mode belonging to the scaffolding teaching theory, which mainly emphasizes the teaching method of cultivating students to promote understanding. Although this teaching mode is very mature in theory, it has not been widely implemented in practical application. In the eyes of many students, the teacher is a teaching tool, and in the eyes of many teachers, the students' performance is their performance. This kind of utilitarian idea will cause teachers and students to have no other emotions besides grades, and as long as things that have nothing to do with improving grades will be ignored. A technique based on technology to improve educational evaluation and learning processes is the interactive teacher evaluation system. It seeks to revolutionize the educational process by encouraging student and teacher participation, individualized learning, and ongoing development. In this way, an interactive teaching evaluation system is constructed to achieve a better effect of helping teachers and students

interact. And scientific emotional education is also a very important education for teachers in the process of teaching students. The survey assesses how well emotional education is incorporated into preschool teaching models by looking at curriculum content, instructional strategies, parental involvement, assessment tactics, training for teachers, resources, and the impact on children's development. It also looks at the difficulties educators face, as well as institutional and policy support for emotional education. Scientific emotional education advocates that the classroom is an interactive stage for teaching and learning, the society is the protagonist of the classroom, a relaxed and harmonious learning atmosphere, and an equal and democratic teacher-student relationship. Emotional education in interactive design education might be the subject of a study that looks at concepts like stress management, ethical concerns, inclusive practices, empathy, feedback, and teamwork. The scope of the survey would be determined by the objectives of the researchers, who would consult scholarly or contemporary literature. Only such a classroom can enable students to learn better, and its concept is the same as that of interactive design teaching.

With the emphasis on teaching methods in education, more and more scholars have studied the subject of interactive design teaching. Among them, Cao A conducted interactive research on situational teaching in

^{*} lideming@jisu.edu.cn

E-mail address: lideming@jisu.edu.cn.

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English classrooms for multimedia networks. (Cao, A. et al., 2017). However, he doesn't give enough examples of situational teaching. Later, Yang B proposed a novel Interactive Teaching Optimizer (ITLO) that optimizes previous teaching and learning methods (Yang, B. et al., 2018). However, the theoretical framework he established in this paper is not very complete. Later, Wanpen I studied the relationship between interactive teaching methods on nursing students' performance and their nursing ability time (INKAEW, W. et al., 2018). However, the questionnaire questions he set up in the text are not comprehensive enough. Chen Li analyzed the application of the interactive teaching mode in the classroom in English teaching. (Chen L. et al., 2021; Xiujuan et al., 2021). However, the data he used in this article are from a long time ago, and it is a bit inappropriate to study it now.

In terms of scientific emotional education, Peng X proposed to use emotion to build a model to test the detailed language content in language, giving new insights into classroom teaching (Peng, X. et al., 2020). However, the theory his model employs is not very appropriate. Jones S outlined how various disciplines are involved in the role of emotion and studied the differences between different disciplinary approaches in pedagogy (Jones, S. et al., 2017). However, he did not give a systematic explanation of emotional education in the text. Kaur, I learn about the benefits of emotional teaching in education and how to incorporate it into effective teaching (Kaur, I. et al., 2019; Zheng et al., 2021). However, the experiments he conducted in the paper did not take into account other influential factors.

The innovation of this paper is to use a different perspective - scientific emotional education to study the construction of an interactive design teaching evaluation system for preschool education in colleges and universities. During the research process, a large amount of data related to this topic was obtained and analyzed simply and effectively. This paper tries to get the best countermeasures for constructing the evaluation system, hoping to construct the interactive design teaching evaluation system perfectly based on scientific emotional education.

This work examines how well an interactive classroom assessment system that uses machine learning techniques and scientific emotion-building performs. The findings demonstrate that building interactive design teaching assessment systems in preschool education is both feasible and successful when utilizing the SVM meta-classifier. Because of its high categorization accuracy rate of 92.1%, interactive teaching evaluation systems under emotional education may be studied in college and university settings.

The rest of the paper is structured as follows: Section 2 describes the construction method of the interactive design teaching evaluation system; Section 3 illustrates the experiments on the construction of the interactive design teaching evaluation system; finally concluded in Section 4.

2. Construction method of interactive design teaching evaluation system

2.1. Interactive design teaching and emotional education

Interactive mode is a teaching mode in which students communicate freely and equally with teachers on learning problems or topics. At this time, teachers' "teaching" and students' "learning" are two-way exchanges and communication many times (Díaz, J. M. et al., 2020; Liu et al., 2020). Influenced by constructivism theory, interactive teaching emphasizes student-centered and realizes pairs of interaction between students, and students through demonstrations, guidance, prompts, explanations, and feedback provided by teachers. This teaching mode is consistent with the current educational philosophy, and the teaching conditions it needs can also be realized by modern technology. Modern technology, such as virtual classrooms, online platforms, and gamification, improves interactive teaching by giving students immersive experiences, immediate feedback, dynamic material display, and hands-on learning opportunities. Under the influence of social interaction, the

interactive teaching mode enables the application of teachers' guiding role and the development of student's independent media system (Díaz, J. M. et al., 2019; Sivaparthipan, C.B. et al., 2022). Through the two-way continuous influence of information interaction and meaning negotiation, the consensus and co-progress between teachers and students are realized, and mutual benefit is finally achieved. The basic framework of this mode is shown in Fig. 1.

As can be seen from Fig. 1, this teaching mode is not a single one, it is a teaching mode formed under the interaction of three parts. and the humanistic environment refers to the teaching atmosphere of the school. By using interactive techniques, multimedia, and a variety of evaluation systems, teachers may create a dynamic learning environment. This promotes active learning, critical thinking, and a supportive learning environment in the classroom while keeping lines of communication open and information current. The interactive relationship refers to the relationship between teaching, between the teaching subject and the environment, and between the elements of the environment. This model becomes an organic whole of teaching because of the existence of these three parts (Martinelli, S. M. et al., 2017; Yang et al., 2022). These three parts are indispensable for this model.

The interactive teaching method has five main characteristics, as shown in Fig. 2.

As shown in Fig. 2, this teaching method mainly emphasizes the cultivation of students' ability to use comprehensive language in the teaching process. Create an English language environment for students in advance and use multimedia to play some pure English and American audio for students, so that students can feel the difference between language and pronunciation, encourage students to listen to more English songs and watch more foreign movies, which will help them improve their foreign language sense (Sujar, A. et al., 2022). The current educational principles generally emphasize the student as the main body, and this teaching method is no exception, which emphasizes that the teacher belongs to a guiding role. Teachers and students belong to an equal relationship. Teachers need to create a lively teaching atmosphere. The most important thing is to focus on the students' cognition in the process of teaching, and not to impose their things on them. Language fluency is also a feature. If the learner constructs a fluent expression, the problems encountered by the students in the learning may not be raised to the teacher, and the teacher will not be able to understand the problem the first time. A student's interest in what they are studying greatly improves their engagement and overall experience. Increased concentration, attention, and critical thinking follow from this. Lessons that are tailored to each student's interests result in individualized learning experiences that encourage long-term engagement and a passion for learning. And the most important communication between people is language. Students have good communication skills, so that no matter what they study, they will be a little handier. When teaching, the subject material selected by the teacher is also very important. It cannot be separated from real life and should be close to students' lives, but it should not be too low-level and naive, causing students to lose confidence (Miller, A. L. et al., 2021).

Emotional education is an integral part of the complete educational process, not something outside the educational reality. Providing people

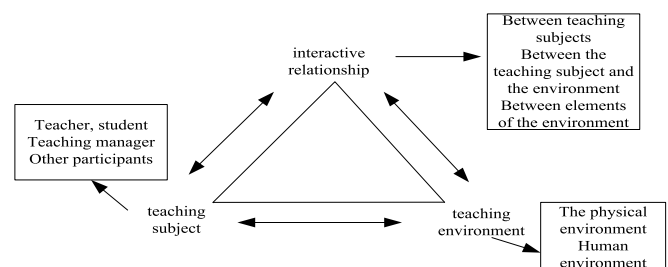


Fig. 1. The basic framework of interactive teaching mode.

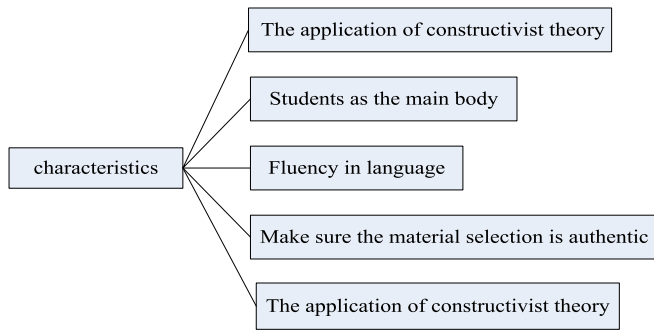


Fig. 2. Characteristics of interactive teaching methods.

with the necessary life skills for success, emotional education is critical for holistic development, mental health, social skills, self-awareness, academic performance, life problems, conflict resolution, empathy, and long-term contentment. In the education process, according to the general policy of emotional education goals, it is important to cultivate students' positive emotions for learning and make them happy to learn. Positive emotional experience of others and things, self-regulation ability, and social-emotional quality are cultivated to form an independent and sound personality and personality, and then become a well-rounded person. Self-regulation is the ability to manage thoughts, emotions, and behaviors effectively, encompassing emotional, cognitive, and behavioral aspects. It enhances decision-making, problem-solving, adaptability, resilience, and autonomy, promoting adaptability, flexibility, and effective stress management. To comprehend and control emotions and relationships, one must possess social-emotional competencies. Empathy, communication, self-awareness, interpersonal skills, and responsible decision-making are important attributes. Early infancy, middle childhood, and adolescence are the developmental stages for these abilities. Students who are physically and mentally sound can often maintain a cheerful and optimistic personality in the learning process (Bakhtiyorovna, K. N. et al., 2021). Its structure diagram is shown in Fig. 3.

The characteristics of emotional education are also very strong. According to the summary, it is found that it mainly has four characteristics, as shown in Fig. 4.

As shown in Fig. 4, in the educational process, students' interest is very important. It is easy for students to increase their concentration on things of interest and to study more seriously (Reisenzein, R. et al., 2018). With interest, students can learn happily and gain knowledge. Only when students experience this kind of success can they be more

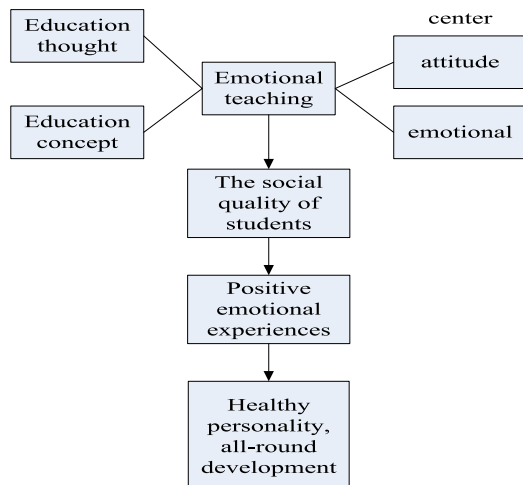


Fig. 3. Structure diagram of emotional education.

engaged in learning and achieve success in learning. Aesthetics is also a very critical point. Aesthetics, creativity, interest-driven learning, and emotional education are important components in helping pupils succeed academically. These components foster resilience, emotional intelligence, and intrinsic drive, setting children up for both academic achievement and lifetime learning. It has nothing to do with people's aesthetics. It mainly focuses on the beauty of reason. Instead of using the classroom as a place to impart knowledge, teachers lead students to experience the scenery of knowledge like a tour guide. Educators guide learners through a "scenery of knowledge," establishing an immersive educational setting that incorporates practical applications and multi-modal teaching techniques. By encouraging curiosity, critical thinking, and a growth mindset, this method seeks to develop lifelong learners. Creativity emphasizes more on teachers' ability to adapt to adaptability, and is no longer sticking to the rules, and will only dryly teach the content of the books. It can be combined with the actual expansion of more knowledge so that students are out of the book but within the scope of book knowledge. And this should also stimulate the creative thinking of students so that they should be flexible and not just blindly study (Tan, P. J. B. et al., 2018).

2.2. Relevant algorithms for teaching evaluation

Teaching evaluation is an activity that uses the theory, method, and technology of educational evaluation to make a comprehensive, of the quality and work performance of teachers according to policies, regulations, and the school's personnel training goals and requirements (Park, H. J. et al., 2018). Teachers may encourage creativity and aesthetics by using technology, promoting open-ended tasks, and fostering an environment for artistic discovery. They foster a lifetime respect for the arts, promote risk-taking, and stimulate a development attitude. With the continuous development of education, the requirements of students, parents, and society for teaching quality are constantly improving. With updated curricula, professional development, technology integration, diversity training, global perspectives, cultural competence, critical thinking, community engagement, life skills, and career preparation, as well as a dynamic educational ecosystem, continuous improvement in education enhances both the quality of instruction and society. Teaching evaluation, as a means of measuring and regulating teaching quality, greatly guarantees the orderly development of various teaching links in colleges and universities. In teaching assessment, the discretization method transforms continuous variables into discrete categories. However, it comes with several drawbacks, including subjectivity, loss of information, influence on model performance, multidimensional data management, and outliers. There are several algorithms for teaching evaluation, and I will list them one by one below.

(1) Association rule algorithm

Its function is to find out the relationship between various attributes in the transaction, as a reference for future transaction prediction or decision-making. For example, in the teaching evaluation database, the teaching manager found that 80% of the professors with doctorate degrees have excellent teaching quality, and it can be inferred that there is a strong correlation between the teaching quality and the teachers' academic titles. Professors with doctorates are excellent teachers because of their broad knowledge, expertise in research, pedagogical training, enthusiasm, adaptability, creativity, mentoring, critical thinking, and dedication to lifelong learning. Therefore, schools can introduce corresponding solutions, allow teachers to participate in training, and fully support teachers to continue their studies, to continuously improve their practical teaching ability. Challenges that include data complexity, sparseness, interpretability, and scalability arise when using association rule algorithms in instructional assessment. To improve interpretability and relevance, addressing these calls for cautious data pretreatment,

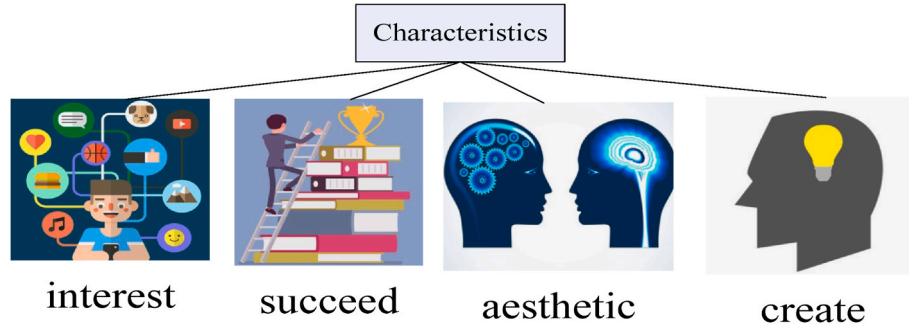


Fig. 4. Characteristics of emotional education.

different techniques, and subject expertise. An important branch of association rules is called data mining, which is the process of finding the technical laws of data and analyzing each data from a large amount of data (Choi, H. et al., 2021). Its system diagram is shown in Fig. 5.

As shown in Fig. 5, the system first performs preprocessing on the original data, then extracts the correct data for analysis, and finally obtains the results for interpretation and evaluation, which is very similar to the evaluation system studied in this paper. Therefore, it is thought that data mining can be used to support the teaching evaluation system (Othman, N. H. et al., 2020).

For people to better understand association rule mining, some basic concepts and definitions related to it are explained as follows.

1) Transactions and items

This definition can be explained in terms of a set of data sets. As shown in Table 1, it is a binary representation table of a set of data. 1 is used to represent the existence of the corresponding column attribute, and 0 to represent the non-existent column attribute.

Each row represents each group of records, called transactions. It can be represented by T, and the unique identifier of each thing can be represented by TID. All records in the entire table can be called the overall transaction set of the database, which can be expressed as:

$$F = \{Y_1, Y_2, Y_3, \dots, Y_N\} \quad (1)$$

Each column in the table represents a data item of the dataset, referred to as an item for short. The number of items that appear in a transaction is called the width of the transaction.

2) Itemset

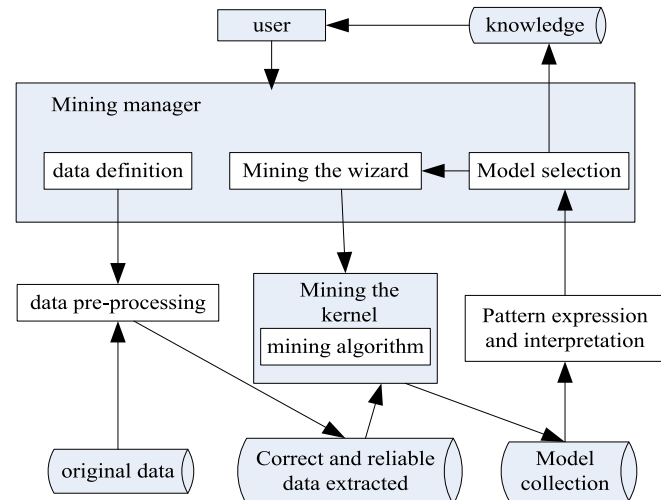


Fig. 5. Data mining system.

Table 1
Binary representation of some data.

TID	a	b	c	d	e	f	g
1	0	0	0	1	0	1	0
2	0	1	0	0	1	0	0
3	1	0	1	0	1	1	1
4	0	1	0	0	0	1	0

In association rules, the collection of data sets is called an item set, which is represented by Y. If an item set Y appears in all transaction sets F, it is called the support number of F to Y, which is represented by Y.C. Then the support S(Y) or probability Q(Y) of the itemset Y can be obtained.

$$S(Y) = \frac{Y.C}{|F|} = Q(Y) \quad (2)$$

Because $S(Y \rightarrow U)$ contains both Y and U, it can be expressed as:

$$S(Y \rightarrow U) = S(Y \cup U) = Q(Y \cup U) \quad (3)$$

Because there are Y and U in $CON(Y \rightarrow U)$, and they also account for the proportion of Y, the formula is expressed as:

$$CON(Y \rightarrow U) = \frac{S(Y \cup U)}{S(Y)} = Q(U|Y) \quad (4)$$

After the elaboration in the previous part, the flow chart of the algorithm can be obtained, as shown in Fig. 6.

As shown in Fig. 6, the mining results of association rules will be directly affected by the process of reasonable discretization. The important link before data mining is data discretization. Data discretization is a preprocessing technique that converts continuous data into discrete intervals or categories, often used to address specific challenges in data analysis or machine learning algorithms that better utilize discrete data. The process helps overcome challenges related to algorithm requirements, noise reduction, and interpretability. A critical preprocessing stage in data mining is data discretization, which enhances interpretability, computing efficiency, and noise reduction, and addresses algorithm assumptions. It manages skewed distributions, keeps continuous data from overfitting, and exposes patterns in the data. A value should correspond to a value in a certain interval, that is, the data should be discretized properly. This work should be completed before mining association rules because it is possible to take continuous data in the original database of index values. Numerical values are discretized into discrete categories or bins to discretize continuous data associated with index values in a database. Effective analyses, algorithms, or applications depend on this process. Various methods are available for discretization, such as equal-width (equal interval), equal-frequency (equal depth), bespoke bins, clustering-based, decision tree-based, and more. Some issues that should be noted should be kept in mind in the mining of association rules because the indicators in the records of the case of taking discrete values are suitable for mining of

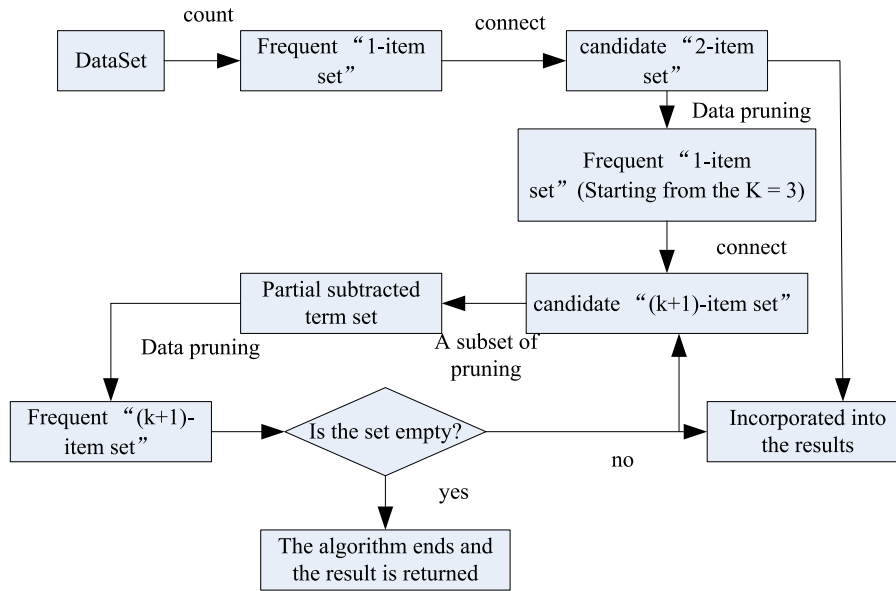


Fig. 6. Algorithm flowchart.

association rules (Barron-Estrada, M. L. et al., 2019).

(2) Support Vector Machine Algorithm

The scope of application of this algorithm is very large, and the applicability is very good. It can still show good classification performance under the condition that the sample data is not large and nonlinear. A reliable and comprehensible option for simulating teaching performance based on scant assessment data is SVM. It reduces overfitting, works well with small datasets, and is efficient in high-dimensional domains. Performance, however, is dependent on the kernel and hyperparameter settings. The SVM algorithm is a supervised learning algorithm, which is widely used in data mining and text classification (Prikhidko, A, et al., 2020). The SVM technique works well in high-dimensional spaces, is resistant to overfitting, accurate in challenging environments, adaptable, memory-efficient, and excellent for sparse data—particularly for text classification—although its suitability for a given task relies on its specific needs.

If dataset $F = \{(C_1, U_1, C_1, U_2), \dots, (C_N, U_N)\}$, the companies used for classification can be represented by the following formula:

$$E^Y C + N = 0 \quad (5)$$

There is an arbitrary point C. If the distance between it and the plane is to be obtained, it can be calculated by the following formula:

$$\rho = \frac{|E^Y C + N|}{||E||} \quad (6)$$

If C_0 is a positive example and is correctly classified, then it satisfies the first inequality. On the contrary, it belongs to the second formula.

$$E^Y C_0 + \geq +1, U_0 = +1 \quad (7)$$

$$E^Y C_0 + \leq -1, U_0 = -1 \quad (8)$$

Such data is generally called a support vector, and the distance between the left and right sides is called the interval, which can be expressed by the following formula:

$$U = \frac{2}{||E||} \quad (9)$$

When the interval is the largest, E and N that meet the definition need to

be found to make U the largest, so it can be gotten:

$$\gamma = \max_{E, N} \frac{2}{||E||} \quad (10)$$

$$D, Y, U_0(E^Y C_0 + N) \geq 1, O = 1, 2, \dots, N \quad (11)$$

This can then be converted to a formula as follows:

$$\gamma = \max_{E, N} \frac{1}{2} ||E||^2 \quad (12)$$

If constraints are added to this formula, the new formula can be obtained as:

$$A(E, N, S) = \frac{1}{2} ||E||^2 + \sum_{O=1}^N S_O (1 - U_0(E^Y C_0 + N)) \quad (13)$$

In this, if A (E, N, S) is set to 0 for E and N, then the formula can be obtained:

$$E = \sum_{O=1}^N S_O U_0 C_0 \quad (14)$$

$$P = \sum_{O=1}^N S_O U_0 \quad (15)$$

3. Experiments on the construction of interactive design teaching evaluation system

3.1. Current situation of teaching

To better understand the development of current preschool teaching models, it is especially aimed at the application of emotional education in interactive design teaching. To give instructional models, ensure communication, adjust for different learning styles, encourage engagement, and provide context, teachers are crucial. Their efficacy demands ongoing professional growth since it has a substantial impact on students' understanding, engagement, and learning achievement. The survey was conducted using multiple dimensions of pedagogical development. And to ensure the authenticity of the data, random sampling was used to investigate the teachers and students of the preschool education departments of three colleges and universities (Li, D. et al., 2022). Personalized learning paths, continuous assessment, data

analytics, instantaneous grading, convenience, gamification, interaction, cooperation, and flexibility are all made possible by the use of technology in the interactive teaching evaluation system in preschool education at universities. The curriculum, instructor approach, classroom atmosphere, developmental phases, cultural variety, availability of resources, parental engagement, evaluation procedures, legal requirements, and inclusion of special needs students all influence the teaching style of preschool education.

The students are listed in Table 2. To ensure the fairness of the sample, The number of people extracted presents a stepped curve (Naftaliev, E. et al., 2013).

The following is the situation table of teachers. Similarly, three school teachers are shown. As shown in Table 3, according to the statistics of teachers' diplomas, each school has selected 20 undergraduate teachers, and these 20 teachers are old teachers who have been teaching for more than one year. 15 graduate teachers and 5 doctoral teachers were also selected (Olsher, S. et al., 2016).

Among them, the content of the questionnaire can be summarized into a table, it can be seen from Table 4 that this survey mainly focuses on interactive teaching. The main purpose is to understand the current level of understanding or application of interactive teaching in preschool education in school, which lays the groundwork for how to better integrate emotional education for a better model update of interactive teaching in the future. The four directions of the questionnaire are given in the table. The first is the basic information, the second is the cognitive situation, and the third is the implementation situation, mainly to understand the specific implementation situation of the current teaching model, and these situations can be better understood by teachers and students. The fourth is the influencing factor. The success or failure of the teaching model will have an influencing factor. It is to find out the influencing factors to make a better evaluation of teaching (Nagari-Haddif, G. et al., 2018).

Then through the survey, data on the awareness of the interactive teaching mode were obtained, as shown in Fig. 7.

As shown in Fig. 7, it can be known that 50% of the teachers in the three schools are familiar with this teaching model, indicating that these teachers have been paying attention to the trend of education trends. They have not stopped at the current situation and have been trying to create a better teaching atmosphere for students and try a better teaching model. More than 20% of teachers have a general understanding of this teaching model. The number of teachers who do not understand this teaching model is less than 20%. Due to concerns with student requirements mismatch, insufficient training, complexity, communication, resistance, time limits, leadership support, and feedback issues, less than 20% of instructors find it difficult to implement instructional models. The reason for not understanding may be that teachers are not very familiar with the concepts, but they may also reflect the same concept in the teaching model. It is also possible that because the popularity of this model is not very high, most teachers still teach according to the teaching objectives of each semester according to the textbook. Schools may also have no rigid requirements for the introduction and innovation of teaching models, and there is no opportunity to understand the significance of these models. As for the students, it can be found that nearly 60% of the students do not understand this model. 60% of students find teaching models difficult because of different communication styles, imprecise instructions, complicated content, disinterest, insufficient resources, pedagogical expertise, and different learning settings. Determining these elements is

Table 2
Number of students in each school in the survey.

university	freshman	sophomore	Junior	senior
A	100	90	80	70
B	100	90	80	70
C	100	90	80	70

Table 3
Basic information about teachers in the survey.

university	undergraduate	Postgraduate	doctor
A	20	15	5
B	20	15	5
C	20	15	5

Table 4
Questionnaire.

The dimension	content	The title
The basic information	Gender, teaching experience, professional title. Online teaching methods and teaching platforms	0
Cognitive situation	Understanding of interactive teaching mode Understanding the role of teachers	1–4 5
implementation	The implementation of interactive teaching The implementation effect of interactive teaching Implementation evaluation of interactive teaching The implementation attitude of interactive teaching	6–10 11 12 13
influence factor	Specific influencing factors	14–21

essential for learning to occur effectively. Maybe no teacher will explain to them about the teaching model in class, and the textbooks that society can access are limited to professional knowledge. Maybe education students will have a little understanding of this model, but it is only limited to textbook knowledge and has not been practiced. The students who have a general understanding of this model are also between 20% and 30%, indicating that these students have heard of the teaching of this model, but they are not very clear about the specific concepts.

After the cognitive survey was conducted, we surveyed the effect of this model on teachers, as shown in Fig. 8.

As can be seen from Fig. 8, most teachers believe that students are still very motivated in class. Among them, nearly 30% of teachers believe that students are highly motivated, indicating that these teachers are very interactive with students in class and have done a very good job in stimulating students' interest. And nearly 50% of teachers feel that learning to perform in the classroom is highly motivated. About 15% of the teachers felt that the student's performance in class was average, which means that the student's performance in the class was not high, which may be related to the student's interest in the course, and also shows that the teaching level of the teachers needs to be improved. About 10% of teachers feel that students are not highly engaged, which shows that students are not very interested in the course. Teachers need to guide and stimulate students' interest so that teachers can better teach. It can be found that the survey results of these three colleges and universities are similar, indicating that the learning situation of students in each school is the same. Then, on the teaching effect, a detailed investigation was also done. From the figure, it can be found that about 25% of teachers think that the effect of this model is particularly good. This shows that these teachers fully understand the true content of the model, and they can make the interaction and communication with students in the classroom based on equality, and effectively mobilize the classroom atmosphere. Nearly 50% of teachers think this model is better, which shows that teachers have recognized the benefits of this model in teaching and started to implement this model. However, nearly 20% of teachers feel that the effect of this model is average, which shows that teachers have not understood the true teaching method of this model, and need to carry out further practical learning. Then there are also a small number of teachers who feel that this model is not good. Maybe these teachers do not understand this model at all, and they are also afraid that accepting new things will affect their teaching progress.

The last investigation is a statistic on the influencing factors of the

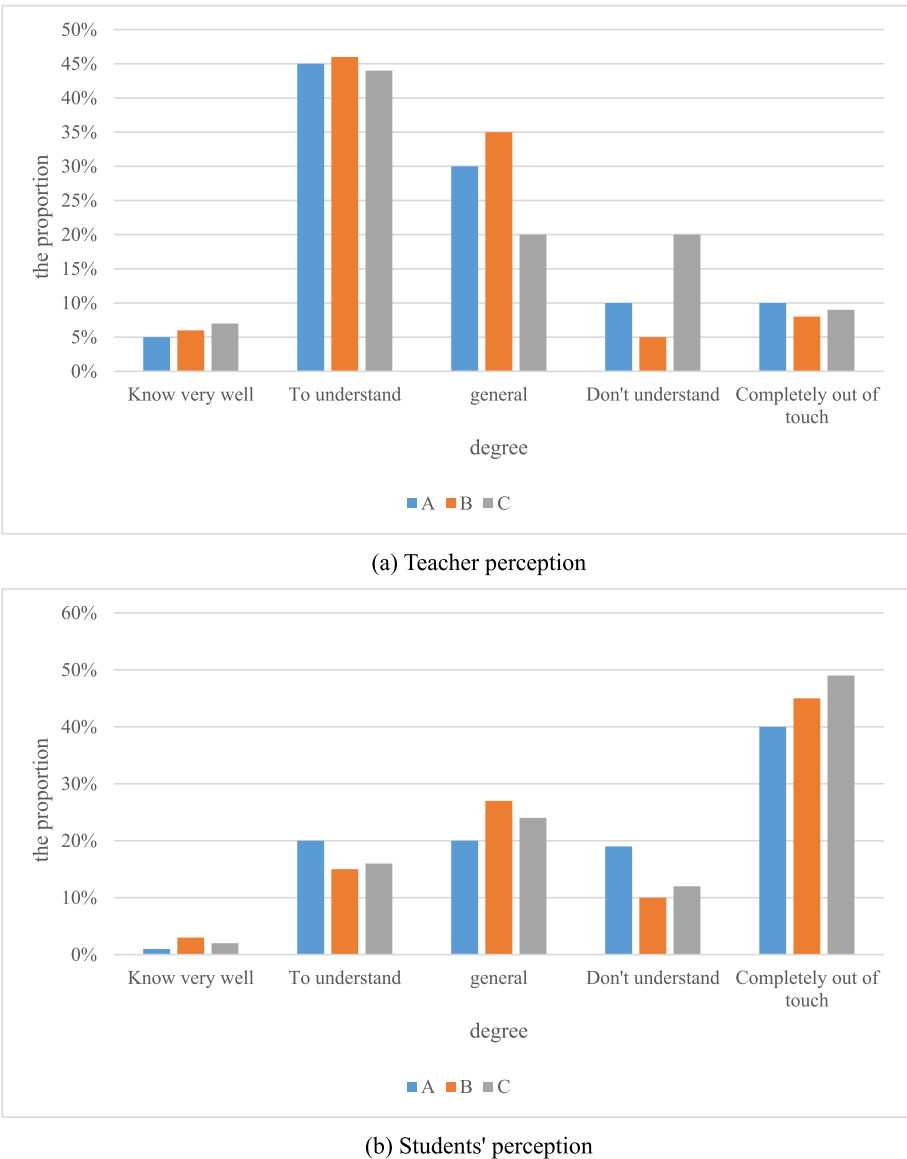


Fig. 7. Teachers and students' knowledge of interactive teaching models.

application of this model.

As can be seen from Table 5, most teachers believe that students' interest in learning has a very large impact on the teaching model, and 36% of them believe that the impact is very large. In the use of multimedia, 54% of teachers believe that teaching will have a great impact. Because multimedia is a way to increase students' interest in learning, it is a good way for students to acquire knowledge outside of textbooks in terms of the use of teaching resources, 72% of teachers feel that it has a great influence. Because the teaching resources are not used properly, in terms of network environment and quality, 45% of teachers feel that it has a great influence. Because many teachings are mainly based on PPT or computer operation, the quality of the network is also very important. As for the content of teaching materials, 54% of teachers think it has a great influence because the content of teaching materials is the most intuitive means to attract students' attention. In terms of teachers' professional level, 45% of teachers believe that this has a great impact because if the teaching level of teachers is not high enough, the creation of the entire teaching model cannot be completed, not to mention the teaching effect. Therefore, in general, there are still many factors that affect the teaching mode of preschool education in school. Colleges and teachers must discuss and study together and strive to find a correct and

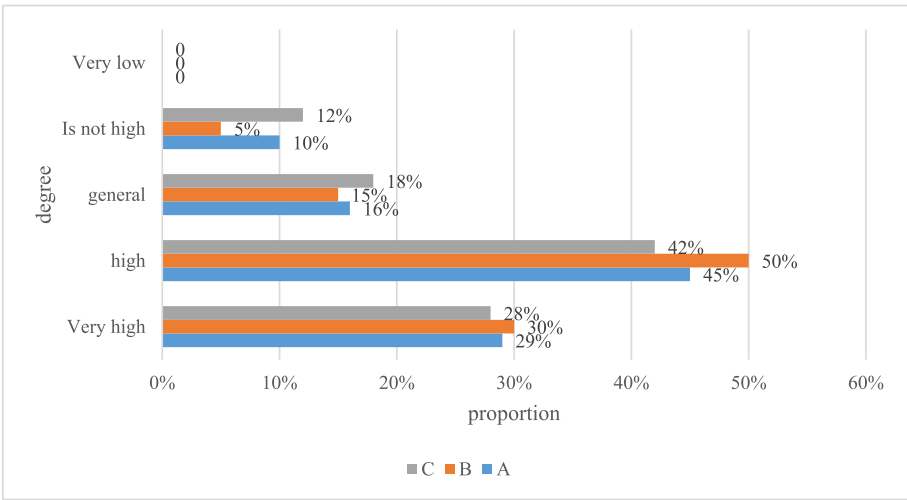
effective interactive teaching method.

3.2. Experiment

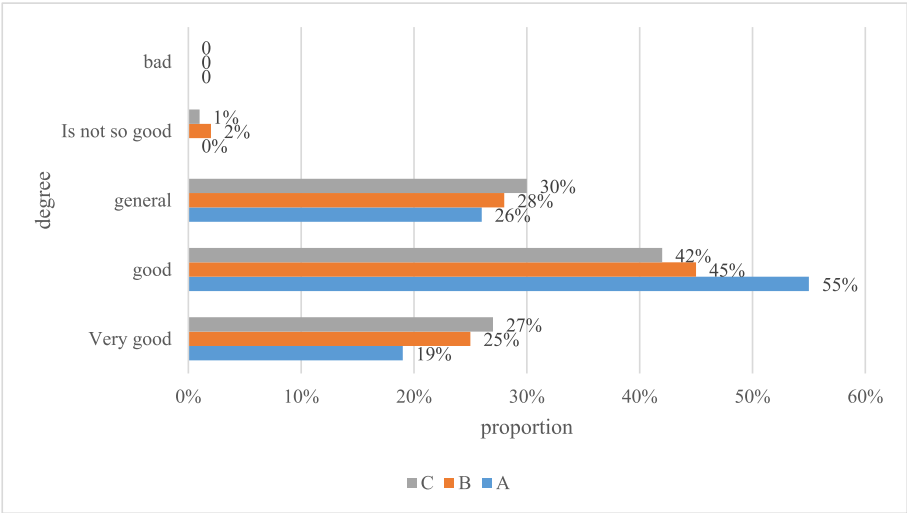
(1) Constructing a hierarchical structure model of evaluation index bids

If the construction of the interactive design teaching evaluation system is to be researched, it is necessary to construct the evaluation index bidding hierarchy model first. When building the model, it should be constructed according to the situation of the school and the teaching quality of the school's preschool education. In general, such models will have first-level indicators and second-level indicators. The first-level indicators are generally w1 to w4, and the second-level indicators are generally w11 to W14, W21 to W24, W31 to W34, and W41 to W43. And they have their corresponding content display, which will not be listed one by one here. Because each school has different teaching quality evaluation standards, it is not easy to make a comprehensive summary.

After the values of these two indicators are obtained, the weights calculated by the formula can be used.



(a) Learning to participate in the enthusiasm of classroom interaction



(b) Teaching effect

Fig. 8. Interactive classroom teaching effect.

Table 5
Factors influencing the application of interactive teaching mode.

The dimension	A great influence	influence	general	Does not affect	Have no effect
Student interest in learning	36%	54%	0	10%	0
Multimedia applications	10%	54%	36%	0	0
Utilization of teaching resources	18%	72%	0	10%	0
Network environment and quality	36%	45%	19%	0	0
The teaching material content	18%	54%	28%	0	0
Professional-level of teachers	45%	36%	19%	0	0

$$\overline{WOK} = \frac{WOK}{\sum_{L=1}^M WLK} \tag{16}$$

However, different research questions also have different evaluation indicators, four indicators can be extended above the statistical results.

The first is the indicator of accuracy, which is mainly to classify the correct samples and obtain the proportion. Its formula is as follows:

$$ACC = \frac{YO + Ym}{YO + GM + GO + YM} \tag{17}$$

The second is the precision metric, whose function is to predict the proportion of normal samples. Its formula is as follows:

$$PRE = \frac{YO}{YO + GO} \tag{18}$$

The third is the recall rate, its function is to predict the proportion of samples whose true class is positive. The formula expression is as follows:

$$RE = \frac{YO}{YO + GM} \tag{19}$$

The last is the G1 value, which is a comprehensive consideration of

the above indicators. If the calculated result is closer to 1, it means that the performance of the model is very good. Its calculation formula is:

$$G1SCO = \frac{2*YO}{2*YO + GM + GO} \quad (20)$$

(2) Performance test

The main research content of this experiment is for the teaching evaluation text. Performance test algorithm selection is influenced by several factors, including model complexity, interpretability, data type, algorithm suitability, computational efficiency, handling missing data, scalability, assumption compliance, prior research, benchmarking, ensemble methods, and domain expertise. This paper intends to use NB, KNN, LR, and SVM algorithms as the meta-learning algorithms in the second layer of the framework, respectively conduct experiments on the algorithm in this paper and statistically analyze the experimental results.

Fig. 9, the ROC curve of the SVM algorithm is closest to the upper left corner of the figure, which can almost cover all the ROC curves of the other three classification algorithms, and its AUC value is higher than the other three classification algorithms. Synthesizing each evaluation index, the classification effect of the meta-classifier is ranked as SVM > LR > KNN > NB. From the above data comparison, the SVM meta-classifier in the algorithm in this paper has achieved the best classification effect. The SVM meta-classifier based on the algorithm in this paper performs well, and the classification accuracy rate reaches 92.1%. This shows that this model is very suitable for studying the construction of an interactive teaching evaluation system under emotional education.

Next, experiments are carried out on the impact of the proposed algorithm on the classification performance. The experimental results are shown in Fig. 10.

As shown in Fig. 10, in terms of performance, the algorithm proposed in this paper significantly improves the performance of all meta-classifiers. Among them, the performance improvement effect of the SVM meta-classifier is the most obvious, the accuracy rate is increased by 1.8 percentage points, and the improvement of the LR meta-classifier is the smallest. Because of its scalability, resistance to outliers, and linear

nature, the LR meta-classifier is less successful than SVM; however, this is dependent on the nature of the issue and the features of the dataset. From the two indicators of accuracy and G1 value, the values of each index of the integrated model are higher than the four primary classifiers, and the accuracy rate exceeds 90%, indicating that the classification performance of the model has been improved and has strong classification ability. The following conclusions can be drawn: the classification model implemented by the algorithm in this paper is feasible and effective for constructing an interactive design teaching evaluation system for preschool education.

4. Conclusions

Through the relevant integration of scientific emotion and machine learning algorithms, the validity of educational problems can be evaluated more effectively (Li, 2022a; Li, 2022b). The interactive design teaching assessment system for preschool education in colleges and universities is analyzed by the study using a data mining method. The algorithm model enhances indications and evaluation accuracy. Nevertheless, it is not practical to conduct a thorough analysis of every facet of the assessment system because of space and example constraints. Research has to be done in-depth with more genuine data.

This study assesses preschool education instruction in colleges and universities and comes to the legitimate conclusion backed by correlation algorithm validity that teacher-student contact improves classroom efficacy and quality. Also, can better improve the school characteristics, teachers' attitudes, and teacher collaboration (Drossel, K. et al., 2017). At the same time, effectively enabling machine learning can integrate the advantages and disadvantages of both, improve the evaluation standards of colleges and basic education, and realize the two-way combination of qualitative and quantitative evaluation. This research trend has been discussed at international conferences on both higher and elementary education (Li and Yang, 2022; Li et al., 2022). With the development of society, the scientific nature of teaching evaluation is becoming more and more popular. It is hoped that in addition to the result evaluation, more process evaluation should be perfected in the evaluation system criteria (Freeman, L. C. et al., 1979), to enrich the interaction between teachers and students or the comprehensive

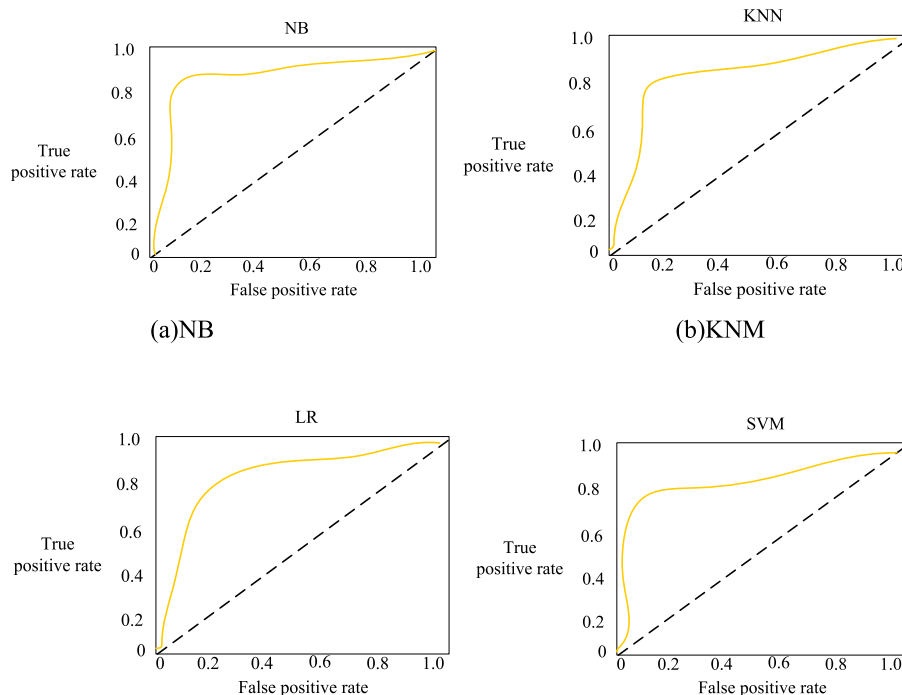


Fig. 9. ROC curve and AUC value of each meta-classifier in this algorithm.

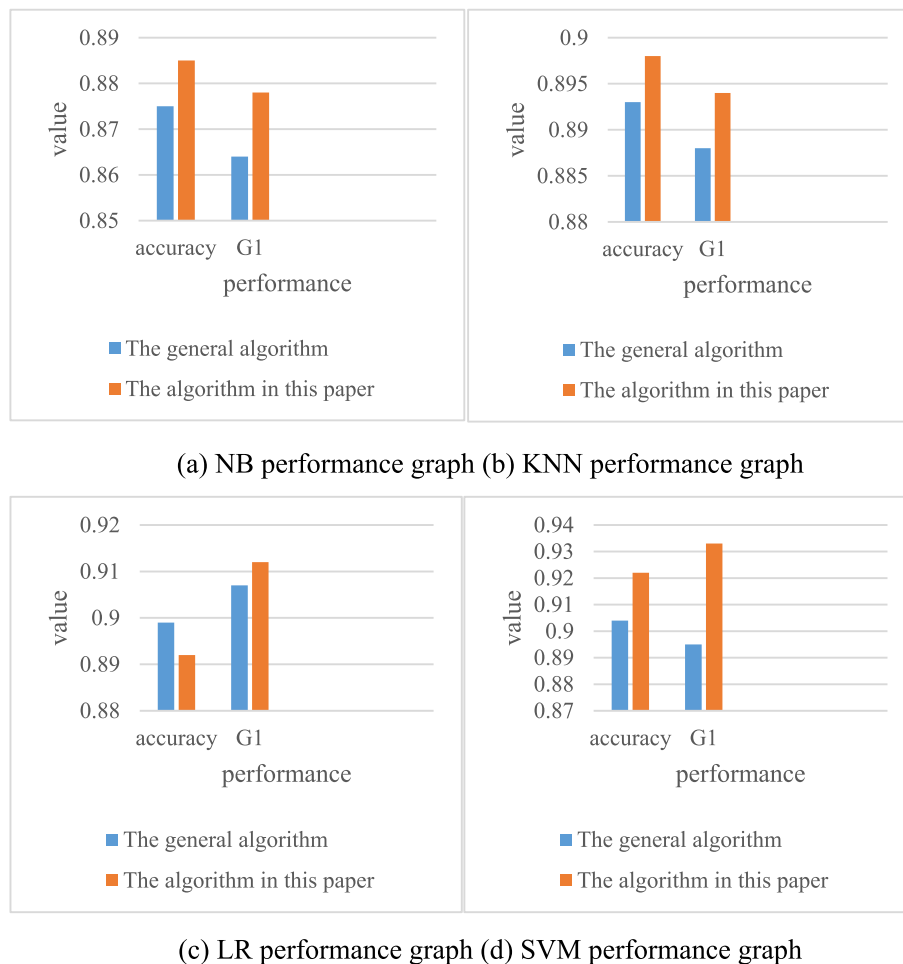


Fig. 10. Comparison of the performance of the two algorithm models.

evaluation criteria of the teaching system will be the further topic of this research in the future. Scientific research is progressing from informatization to intelligence thanks to the ongoing development of educational technology and digital transformation. The interactive technology of artificial intelligence (Zheng, W. et al., 2023) continues to dig deeper, deep learning (Mohamed, A. A. et al., 2023), machine learning (Shafiezadeh, S. et al., 2023) and other algorithms will be more widely applied in pedagogy and other comprehensive disciplines to develop new data testing and experiment mining, which has important research value to promote the in-depth research of interdisciplinary integration. Using machine learning techniques, the project seeks to develop an interactive teaching assessment system for preschool instruction in higher education. Personalization, data integration, ongoing feedback, and ethical issues are among the project's future goals.

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Code availability

Not applicable.

CRedit authorship contribution statement

Deming Li: Formal analysis, Data curation.

Declaration of competing interest

The author declares that there are no conflicts of interest. The author read and approved the manuscript for submission.

Data availability

The authors do not have permission to share data.

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Update

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Subsequent to the acceptance of these special issue papers by the Guest Editors, concerns have been raised about the integrity and rigor of the peer-review process. The journal is currently investigating the

concerns. If deemed necessary following this assessment, readers will be further updated in accordance with Elsevier and Committee on Publication Ethics best practices.

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